

Further Statistics 1

Notes · Edexcel A Level Further Mathematics — Further Statistics 1 · Chapter 5: Central Limit Theorem

Chapter 5: The Central Limit Theorem

Why this chapter matters

So far, whenever you wanted the distribution of a sample mean to be Normal, you needed the underlying population itself to be Normal. The Central Limit Theorem (CLT) removes that restriction: it tells you that, for a large enough sample, the distribution of the sample mean is *approximately* Normal regardless of the shape of the parent population. This is one of the most powerful and famous results in all of statistics, and it is what allows Normal-based methods to be applied so widely in the real world.

Concept: statement of the Central Limit Theorem

If $X_1, X_2, \dots, X_n$ are independent observations from any distribution (not necessarily Normal) with mean μ and variance σ^2 , then for sufficiently large n , the sample mean $\bar{X}$ is approximately Normally distributed:

$$\bar{X} \sim N(\mu, \sigma^2/n) \text{ approximately, for large } n.$$

Equivalently, the sample total $T = X_1 + X_2 + \dots + X_n$ is approximately Normal: $T \sim N(n\mu, n\sigma^2)$.

Crucially: the *exact* mean and variance of $\bar{X}$ (namely μ and σ^2/n) hold for any n and any parent distribution — that part is not new. What the CLT adds is that the *shape* of the distribution of $\bar{X}$ becomes approximately Normal as n increases, even when the parent population is heavily skewed, discrete, or otherwise non-Normal.

Why is this surprising? Picture rolling an ordinary die: the distribution of a single roll is flat (uniform) — every value 1 to 6 equally likely, nothing Normal about it at all. Now roll it 30 times and look at the *mean* of those 30 rolls. Averaging washes out the individual roll-to-roll unevenness: a run of high rolls tends to be balanced out by lower ones elsewhere, so the mean clusters tightly around 3.5 in a bell-shaped way, however lopsided or blocky the original die's distribution was. The CLT says this bell shape for the mean is essentially universal — it doesn't care whether the parent distribution is uniform, wildly skewed, or even discrete, only that you're averaging enough independent copies of it.

Key vocabulary

Term	Meaning
Parent population	The original distribution from which individual observations are drawn
Sample mean, $\bar{X}$	The mean of a random sample of size n
“Large enough” n	In exam contexts, $n \geq 30$ is generally treated as large enough for the CLT approximation to be used confidently, though the required size depends on how non-Normal the parent distribution is

Exact distribution	The true distribution of $\bar{X}$, only Normal automatically if the parent population is itself Normal
Approximate distribution (CLT)	The Normal distribution used to approximate $\bar{X}$ when the parent is non-Normal, valid for large n

Key formula: distribution of the sample mean

KEY FORMULA	
Situation	Distribution of $\bar{X}$
Parent population is Normal, any n	Exactly $N(\mu, \sigma^2/n)$
Parent population is non-Normal, n large	Approximately $N(\mu, \sigma^2/n)$ by the CLT

Worked example 1: applying the CLT to a non-Normal parent population

EXAMPLE 5.1

Question. The time, in minutes, that customers spend browsing a website has mean 8 and standard deviation 5. The distribution is heavily skewed and not Normal. A random sample of 50 browsing times is taken. Find the approximate probability that the sample mean browsing time exceeds 9 minutes.

Step 1. Since $n = 50$ is large, apply the Central Limit Theorem: $\bar{X} \sim N(8, 5^2/50)$ approximately, i.e. $\bar{X} \sim N(8, 0.5)$ approximately. **(M1)**

Step 2. Standardise: $Z = (9 - 8)/\sqrt{0.5} = 1/0.7071 \approx 1.4142$. **(M1A1)**

Step 3. $P(\bar{X} > 9) = P(Z > 1.4142) = 1 - \Phi(1.4142) \approx 1 - 0.9214 = 0.0786$. **(M1A1)**

Answer: 0.0786 (approximately)

EXAM TIP

The step most often skipped is explicitly justifying the use of the CLT — write “ $n = 50$ is large, so by the Central Limit Theorem...” as a mandatory sentence, because mark schemes award a method mark specifically for invoking the CLT correctly (not just for getting the right numbers). Also remember that we divide the variance (not the standard deviation) by n — confusing σ^2/n with σ/n is the most common numerical slip in this chapter.

Concept: CLT with sample totals

Some questions phrase things in terms of the sample total rather than the sample mean. Since $T = n\bar{X}$, and $\bar{X} \sim N(\mu, \sigma^2/n)$ approximately, it follows that $T \sim N(n\mu, n\sigma^2)$ approximately (using the linear transformation results from Chapter 1: $E(n\bar{X}) = n\mu$ and $\text{Var}(n\bar{X}) = n^2 \times \sigma^2/n = n\sigma^2$).

Worked example 2: CLT with a sample total
EXAMPLE 5.2

Question. The weight of a randomly chosen apple from a large orchard has mean 120 g and standard deviation 18 g (the distribution of weights is not Normal). A random sample of 40 apples is selected. Find the approximate probability that the total weight of the sample exceeds 5000 g.

Step 1. Let T be the total weight of 40 apples. By the CLT ($n = 40$ is large), $T \sim N(40 \times 120, 40 \times 18^2) = N(4800, 12960)$ approximately. **(M1A1)**

Step 2. Standardise: $Z = (5000 - 4800)/\sqrt{12960} = 200/113.84 \approx 1.7566$. **(M1A1)**

Step 3. $P(T > 5000) = P(Z > 1.7566) = 1 - \Phi(1.7566) \approx 1 - 0.9605 = 0.0395$. **(M1A1)**

Answer: 0.0395 (approximately)

EXAM TIP

It is tempting to standardise a total using σ (18) instead of $\sqrt{n\sigma^2}$. A helpful check: remember that the variance of a *sum* of n independent variables is n times the variance of one variable — this is not new statistics, it's the same additive variance rule you used for sums of independent random variables, just applied at scale. Linking it back to that familiar rule (rather than presenting a new formula to memorise) reduces errors significantly.

Concept: when the CLT is (and isn't) needed

A common exam trap is a question where the parent population is already stated to be Normal — in that case, $\bar{X}$ is *exactly* Normal for any sample size, and invoking the CLT is unnecessary (though harmless to mention that the result also holds exactly here). The CLT is only doing genuine work when the parent population is non-Normal, or its shape is unknown, and the sample size is large.

Note that “non-Normal” includes the discrete distributions from earlier chapters: the parent population could just as well be Binomial or Poisson (e.g. “a random sample of 40 households, each recording their number of Poisson-distributed accidents last year”) — the CLT applies exactly the same way, since it only cares about the parent's mean and variance, not whether it is discrete or continuous.

Worked example 3: recognising when the CLT is not required
EXAMPLE 5.3

Question. The heights of adult men in a population are Normally distributed with mean 178 cm and standard deviation 7 cm. A random sample of 4 men is taken. Find the probability that their mean height exceeds 180 cm.

Step 1. Since the parent population is already Normal, $\bar{X} \sim N(178, 7^2/4) = N(178, 12.25)$ exactly — no CLT approximation is needed, and note $n = 4$ is small, so the CLT could not have been used here anyway if the parent were non-Normal. **(B1)**

Step 2. Standardise: $Z = (180 - 178)/\sqrt{12.25} = 2/3.5 \approx 0.5714$. **(M1A1)**

Step 3. $P(\bar{X} > 180) = P(Z > 0.5714) = 1 - \Phi(0.5714) \approx 1 - 0.7161 = 0.2839$. **(M1A1)**

Answer: 0.284 (3 s.f.)

EXAM TIP

This example exists precisely to catch out anyone who reflexively writes “by the CLT” on every question involving a sample mean. Ask yourself: “is the parent population Normal? If yes, do we need the CLT at all?” If you understand the distinction, you will recognise that the exact result already applies, and that a large sample would only be required if the population were non-Normal.

Chapter summary

Idea	Key point
Central Limit Theorem	For large n , $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$, whatever the shape of the parent population
“Large enough” n	$n \geq 30$ is a standard exam rule of thumb
Sample total	$T = n\bar{X}$ is approximately $N(n\mu, n\sigma^2)$
Normal parent population	$\bar{X}$ is exactly Normal for ANY n — the CLT is not needed (though its conclusion still holds)
Non-Normal parent population	The CLT is essential and must be explicitly stated to justify using a Normal approximation

Common exam mistakes

- Dividing by n instead of $\sqrt{n}$ when standardising $\bar{X}$ (confusing standard deviation of $\bar{X}$ with variance of $\bar{X}$).
- Using σ^2 instead of $n\sigma^2$ for the variance of the sample total T .
- Failing to explicitly state “by the Central Limit Theorem, since n is large” as a justification, losing an easy method mark.
- Invoking the CLT unnecessarily (or incorrectly assuming a Normal result is only approximate) when the parent population is already stated to be Normal.
- Treating a small sample from a non-Normal population as approximately Normal when n is not large enough for the CLT to apply.
- Thinking the CLT means *individual* observations become Normally distributed — it does not. The parent population keeps its original (possibly very non-Normal) shape; only the sample mean $\bar{X}$ (or total T) becomes approximately Normal.

You should now be able to

- State the Central Limit Theorem precisely, including the conditions under which it applies.
- Use the CLT to find approximate probabilities involving a sample mean from a non-Normal population.

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- Use the CLT to find approximate probabilities involving a sample total.
 - Recognise when a Normal distribution for $\bar{X}$ is exact (parent population Normal) versus when it depends on the CLT approximation (parent population non-Normal, n large).